

Lab 3: Testing Classes with Dependencies

Objective

Learn how to test classes that depend on other services by using mocking frameworks.

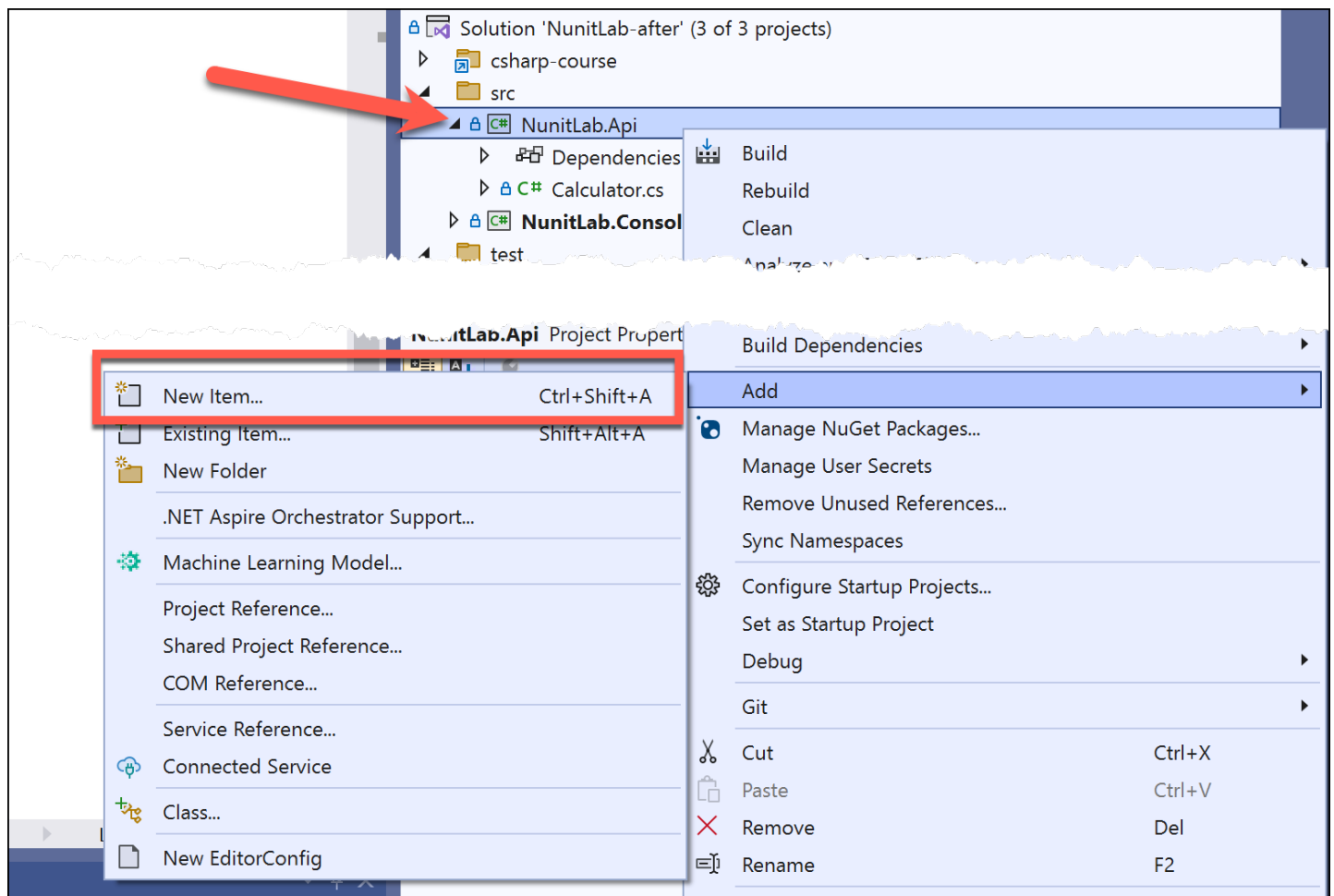
Prerequisites

- Completion of **Lab 2** or familiarity with basic NUnit tests and exception handling.
- NuGet package for Moq installed in the `NunitLab.UnitTests` project.

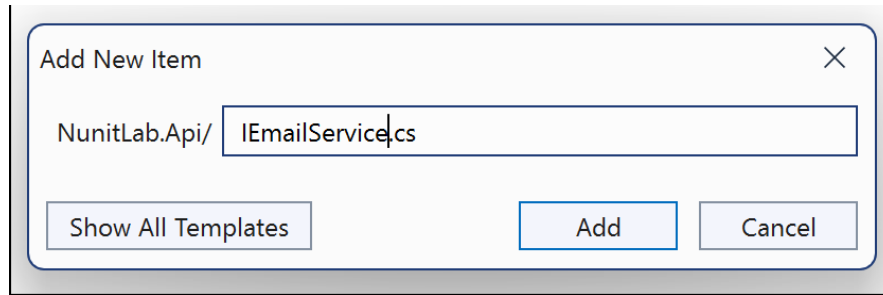
Instructions

Step 1: Create the `IEmailService` and `OrderProcessor` Classes

1. In the `NunitLab.Api` project, create a new interface `IEmailService.cs`. To do this, right-click on the **Nunit.API** project in **Solution Explorer**, go to **Add > New Item...**



2. You should now see the **Add New Item** dialog. Type `IEmailService.cs` in the box and click the **Add** button



3. You should now see the new interface file - **IEmailService.cs**

4. Modify **IEmailService** to look like the code below:

```
namespace NunitLab.Api;

public interface IEmailService
{
    void SendEmail(string recipient, string subject, string body);
}
```



5. Create a new class `OrderProcessor.cs` that uses `IEmailService`:

```
namespace NunitLab.Api;

public class OrderProcessor
{
    private readonly IEmailService _emailService;

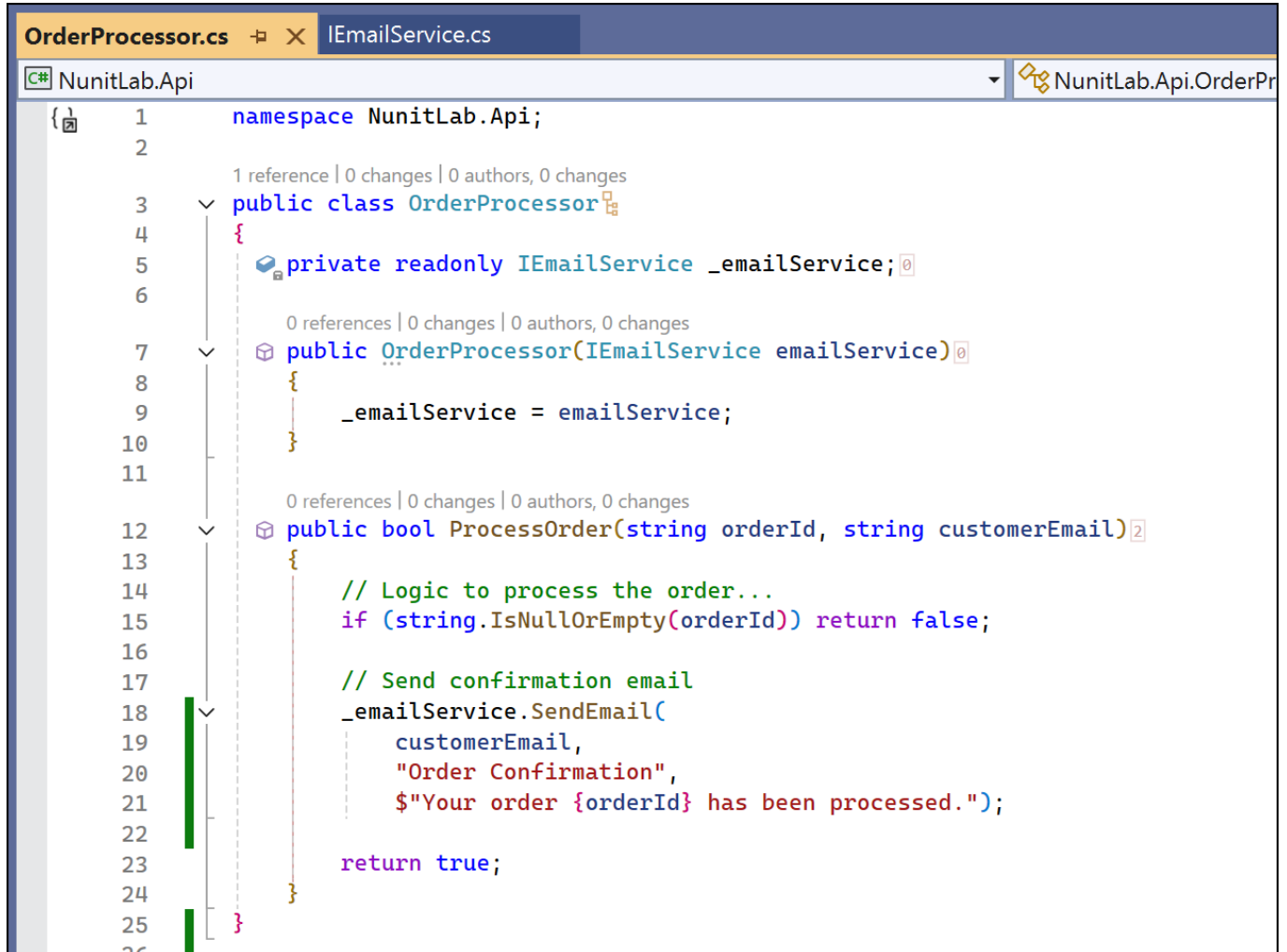
    public OrderProcessor(IEmailService emailService)
    {
        _emailService = emailService;
    }

    public bool ProcessOrder(string orderId, string customerEmail)
    {
        // Logic to process the order...
        if (string.IsNullOrEmpty(orderId)) return false;
    }
}
```

```

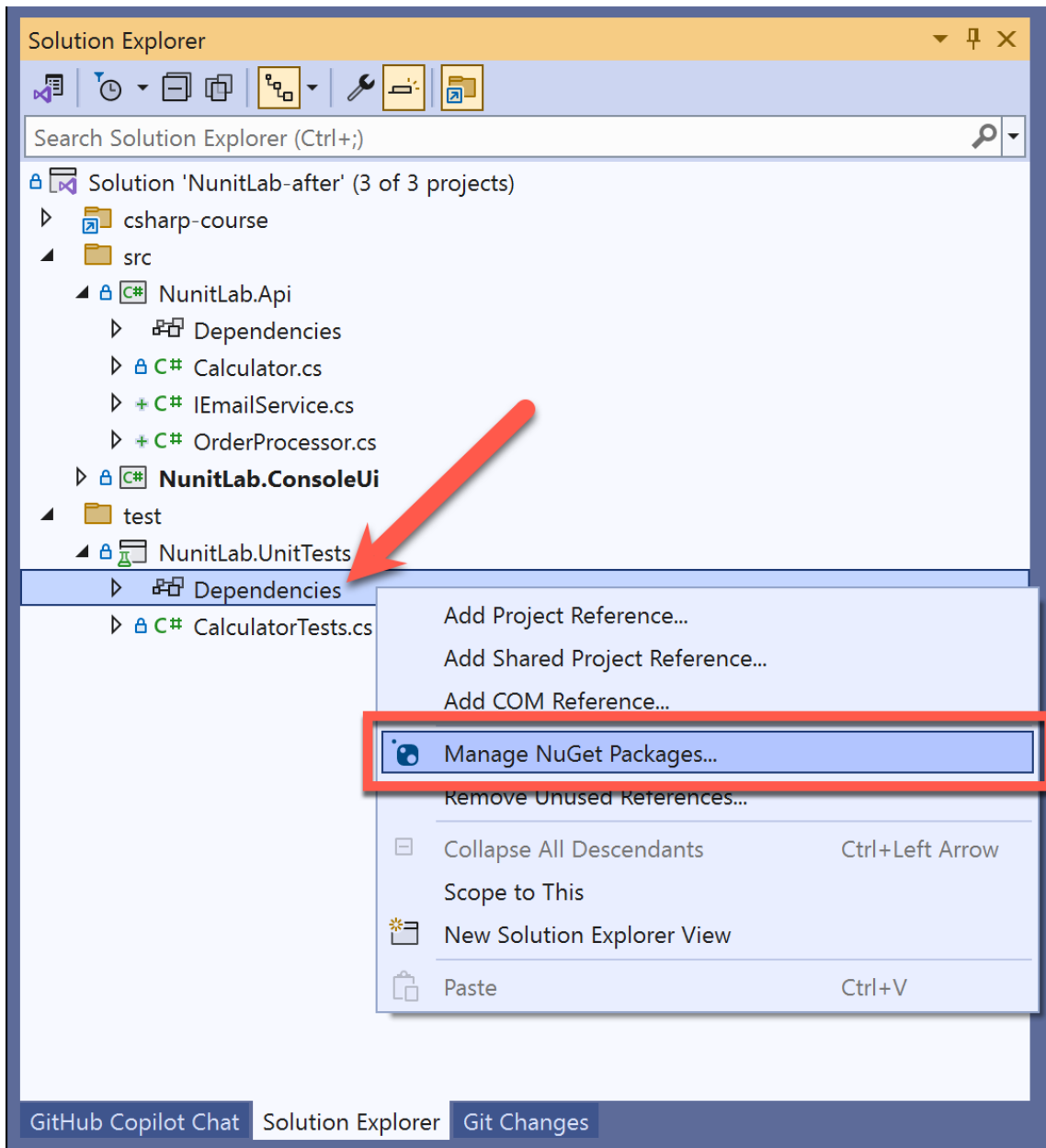
        // Send confirmation email
        _emailService.SendEmail(customerEmail, "Order Confirmation", $"Your order
{orderId} has been processed.");
        return true;
    }
}

```

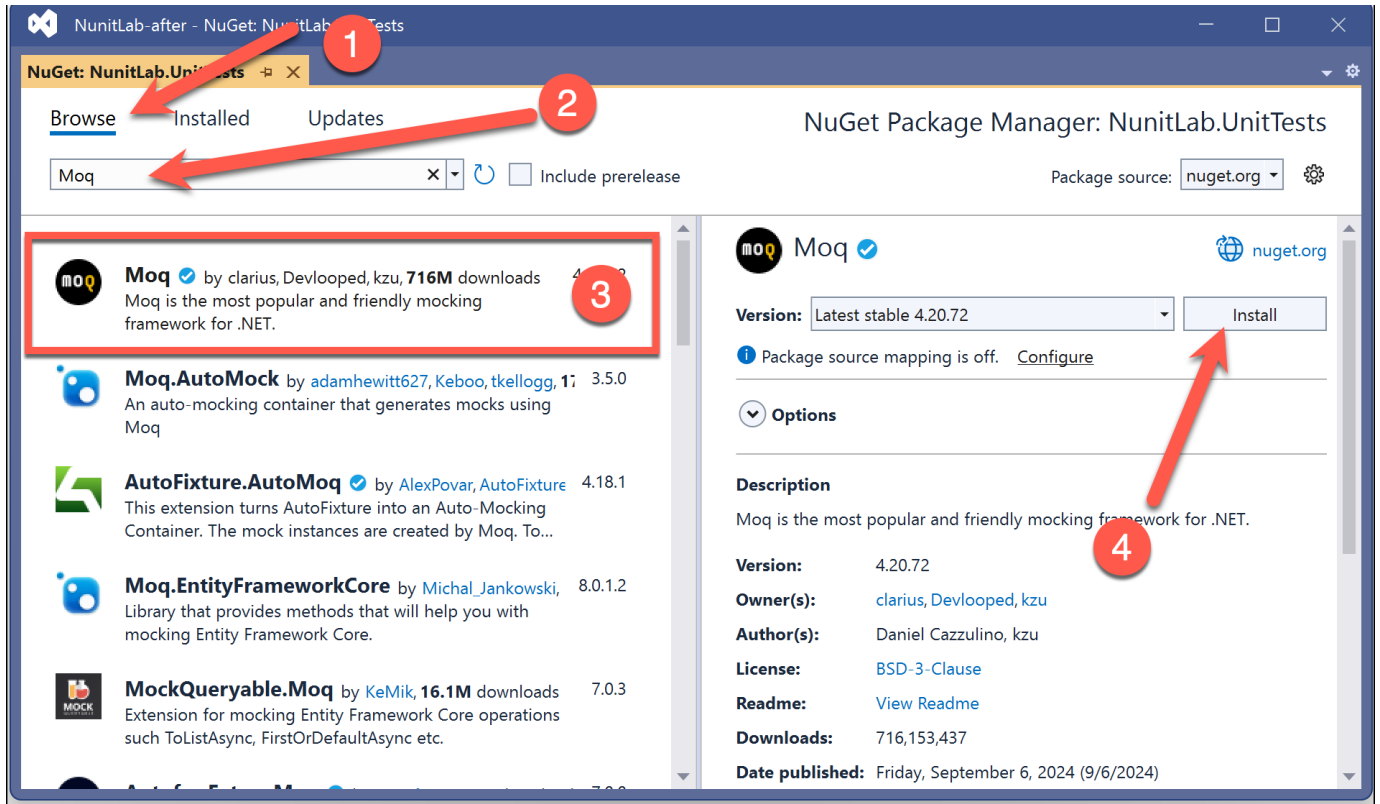


Step 2: Add Tests for OrderProcessor

1. In the `NunitLab.UnitTests` project, install the Moq NuGet package:
 - Expand the project so that you can see the **Dependencies** node
 - Right-click on the **Dependencies** node, choose **Manage NuGet Packages**



- Select the **Browse** tab
- Type **moq** in the search box
- Select **Moq** from the results
- Click the **Install** button to install Moq package into the unit test project



2. Create a new test class `OrderProcessorTests.cs`:

```
using Moq;
using NunitLab.Api;

namespace NunitLab.UnitTests;

[TestFixture]
public class OrderProcessorTests
{
    private OrderProcessor? _systemUnderTest;
    private Mock<IEmailService>? _emailServiceMock;

    public OrderProcessor SystemUnderTest
    {
        get
        {
            if (_systemUnderTest == null)
            {
                _systemUnderTest = new OrderProcessor(EmailServiceMock.Object);
            }

            Assert.That(_systemUnderTest, Is.Not.Null);
            Assert.That(_emailServiceMock, Is.Not.Null);
        }
    }
}
```

```

        return _systemUnderTest;
    }
}

public Mock<IEmailService> EmailServiceMock
{
    get
    {
        if (_emailServiceMock == null)
        {
            _emailServiceMock = new Mock<IEmailService>();
        }

        Assert.That(_emailServiceMock, Is.Not.Null);

        return _emailServiceMock;
    }
}

[SetUp]
public void SetUp()
{
    _systemUnderTest = null;
    _emailServiceMock = null;
}

[Test]
public void ProcessOrder_WhenOrderIdIsValid_ShouldSendEmail()
{
    // Arrange
    string orderId = "12345";
    string customerEmail = "customer@example.com";

    // Act
    var actual = SystemUnderTest.ProcessOrder(orderId, customerEmail);

    // Assert
    Assert.That(actual, Is.True);

    EmailServiceMock.Verify(
        x => x.SendEmail(
            customerEmail,
            "Order Confirmation",
            $"Your order {orderId} has been processed."),
        Times.Once);
}

[Test]
public void ProcessOrder_WhenOrderIdIsInvalid_ShouldNotSendEmail()

```

```

{
    // Arrange
    string orderId = "";
    string customerEmail = "customer@example.com";

    // Act
    var actual = SystemUnderTest.ProcessOrder(orderId, customerEmail);

    // Assert
    Assert.That(actual, Is.False);
    EmailServiceMock.Verify(x => x.SendEmail(It.IsAny<string>(), It.IsAny<string>()),
    It.IsAny<string>()), Times.Never);
}
}

```

```

OrderProcessorTests.cs CalculatorTests.cs OrderProcessor.cs IEmailService.cs
NunitLab.UnitTests NunitLab.UnitTests.OrderProcessorTests
8 public class OrderProcessorTests
53 | 0 references | 0 changes | 0 authors, 0 changes
54 | public void ProcessOrder_WhenOrderIdIsValid_ShouldSendEmail()
55 | {
56 |     // Arrange
57 |     string orderId = "12345";
58 |     string customerEmail = "customer@example.com";
59 |
60 |     // Act
61 |     var actual = SystemUnderTest.ProcessOrder(orderId, customerEmail);
62 |
63 |     // Assert
64 |     Assert.That(actual, Is.True);
65 |
66 |     EmailServiceMock.Verify(
67 |         x => x.SendEmail(
68 |             customerEmail,
69 |             "Order Confirmation",
70 |             $"Your order {orderId} has been processed."),
71 |         Times.Once);
72 | }
73 |
74 | [Test]
75 | | 0 references | 0 changes | 0 authors, 0 changes
76 | | public void ProcessOrder_WhenOrderIdIsInvalid_ShouldNotSendEmail()
77 | | {
78 | |     // Arrange
79 | |     string orderId = "";
80 | |     string customerEmail = "customer@example.com";
81 | |
82 | |     // Act
83 | |     var actual = SystemUnderTest.ProcessOrder(orderId, customerEmail);
84 | |
85 | |     // Assert
86 | |     Assert.That(actual, Is.False);
87 | |     EmailServiceMock.Verify(x => x.SendEmail(It.IsAny<string>(), It.IsAny<string>(), It.IsAny<string>()), Times.Never);
88 | | }
89 | }

```

Step 3: Run and Verify Tests

1. Open the **Test Explorer** in Visual Studio.
2. Run all tests and verify that:
 - The email is sent only when the order ID is valid.
 - No email is sent for invalid order IDs.

Test Explorer

Test run finished: 10 Tests (10 Passed, 0 Failed, 0 Skipped) 0 Warnings 0 Errors

Test	Durat...	Traits
✓ NunitLab.UnitTests (10)	215 ms	
✓ NunitLab.UnitTests (10)	215 ms	
✓ CalculatorTests (8)	27 ms	
✓ Add_ShouldReturnCorrectSum	25 ms	
✓ Divide_WhenDividingByZero_ShouldThrowDivideB...	2 ms	
✓ Multiply_ShouldReturnCorrectProduct	< 1 ms	
✓ Divide_ShouldReturnCorrectQuotient (4)	< 1 ms	
✓ Subtract_ShouldReturnCorrectDifference	< 1 ms	
✓ OrderProcessorTests (2)	188 ms	
✓ ProcessOrder_WhenOrderIdIsInvalid_ShouldNotSe...	182 ms	
✓ ProcessOrder_WhenOrderIdIsValid_ShouldSendEm...	6 ms	